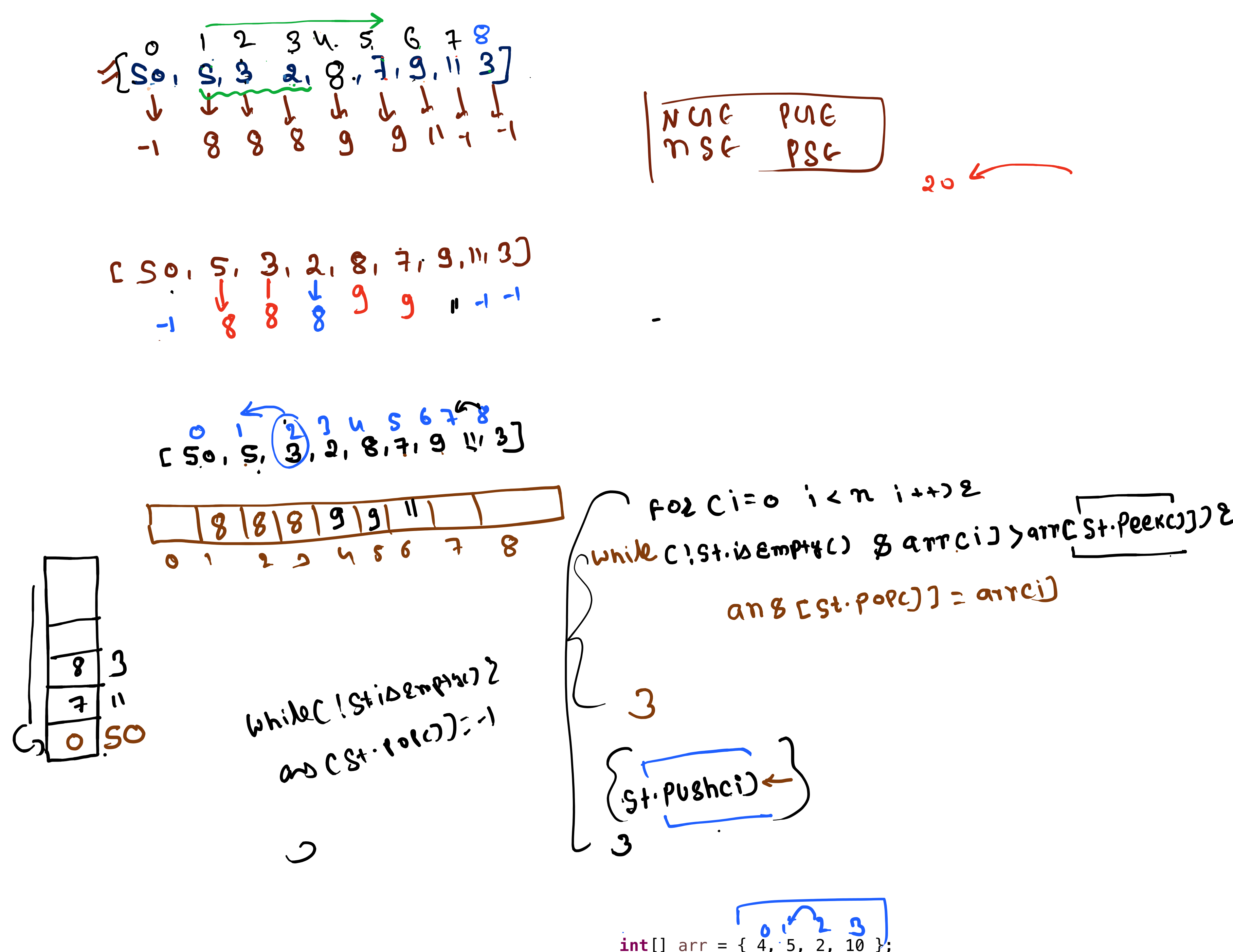
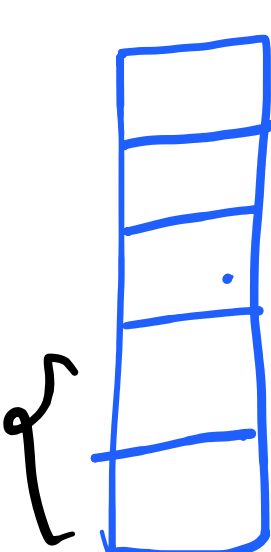
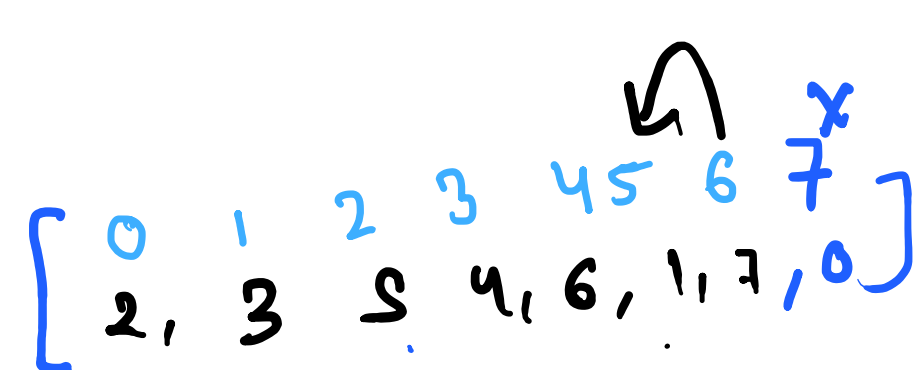
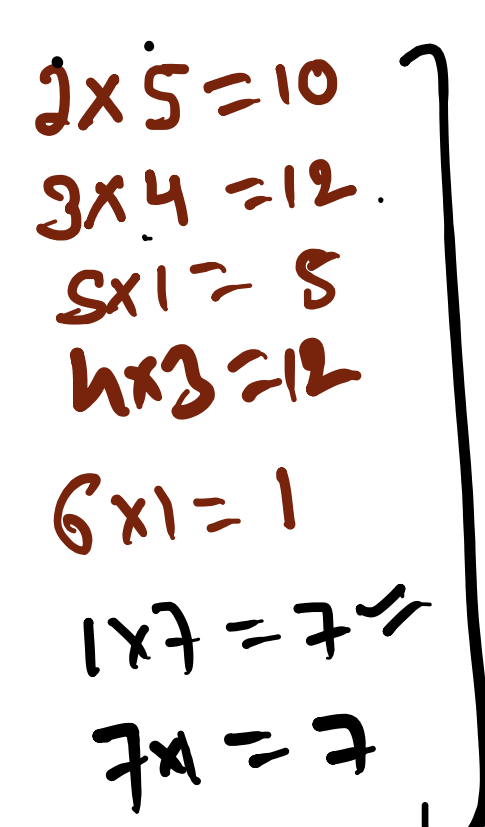
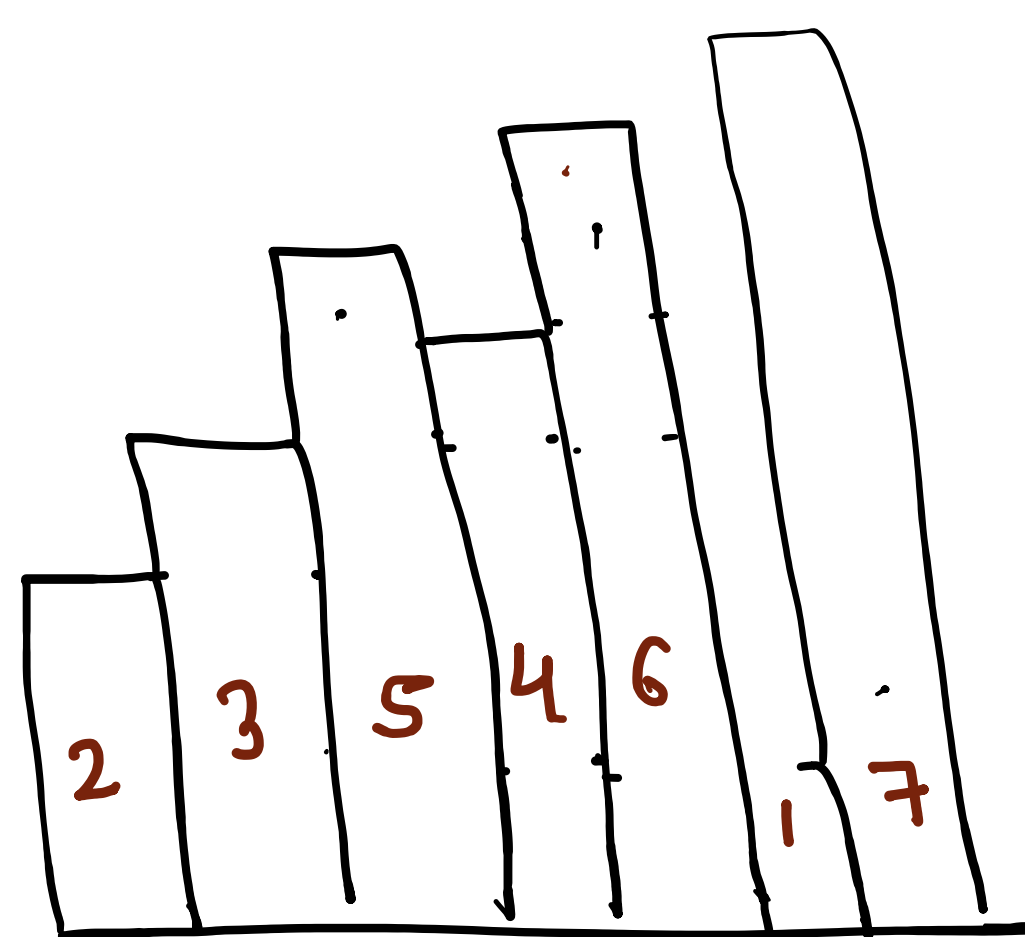
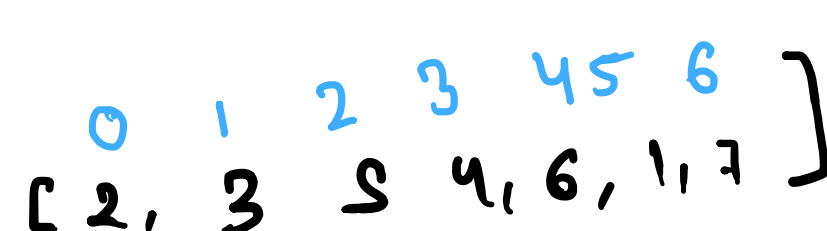
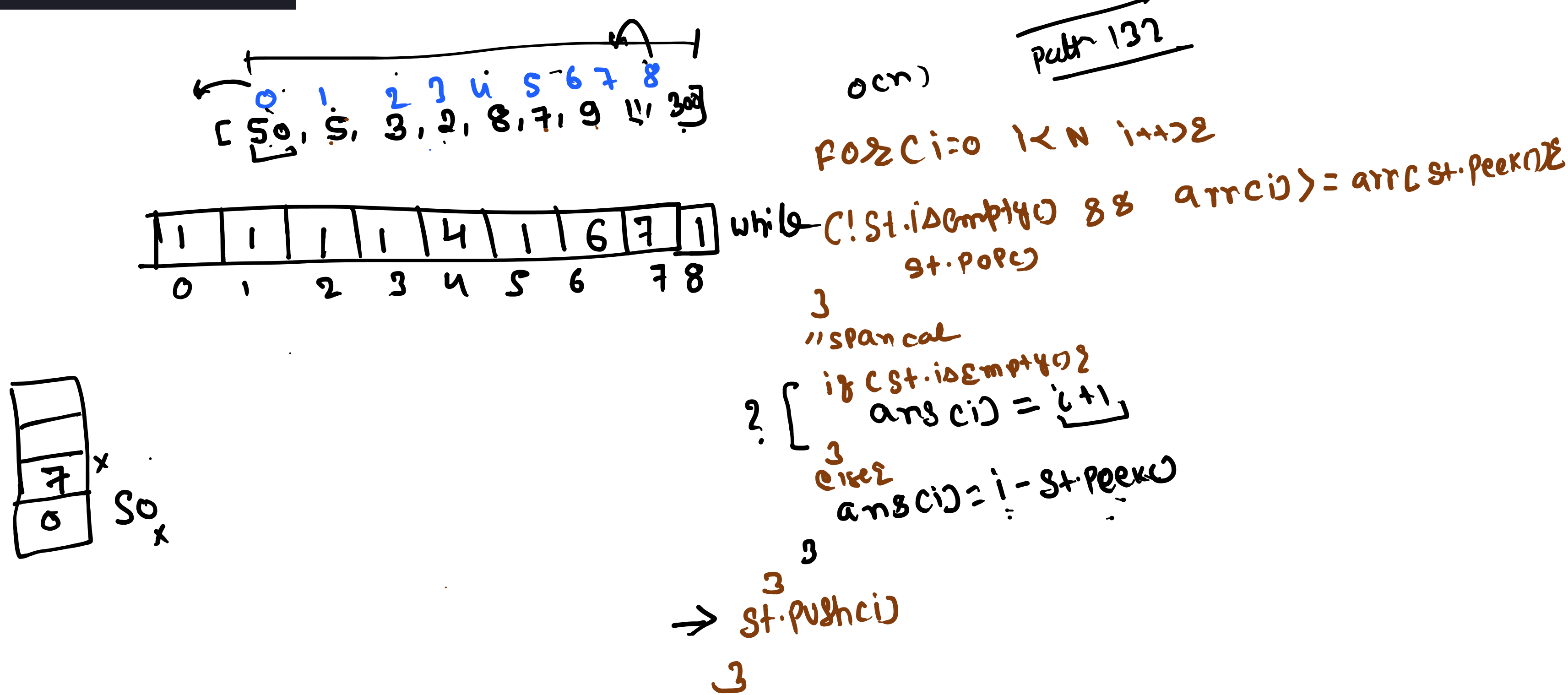
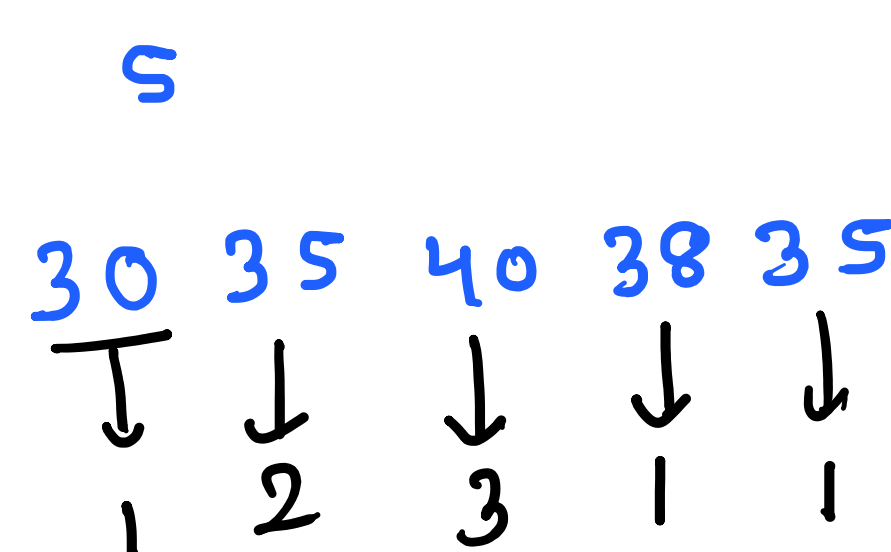
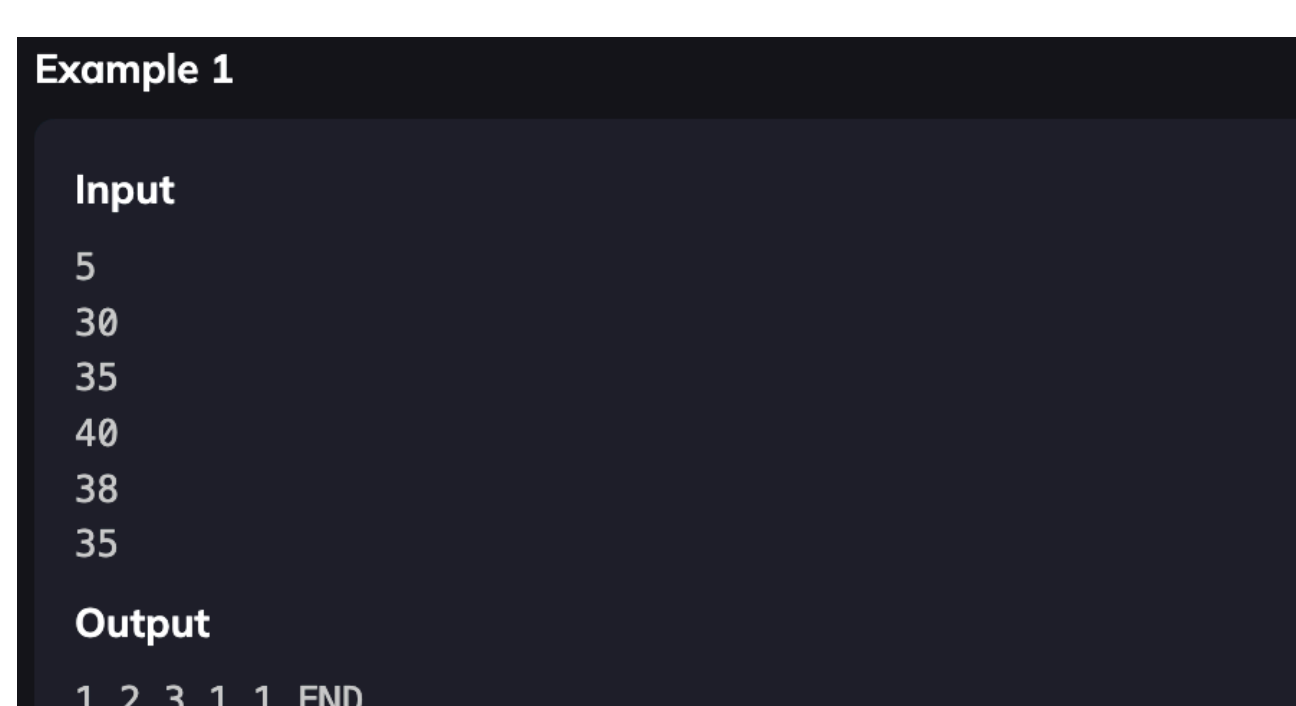
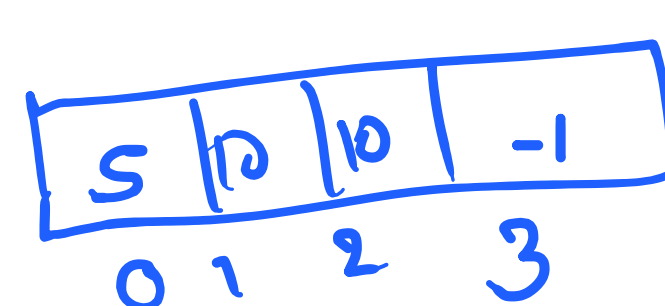
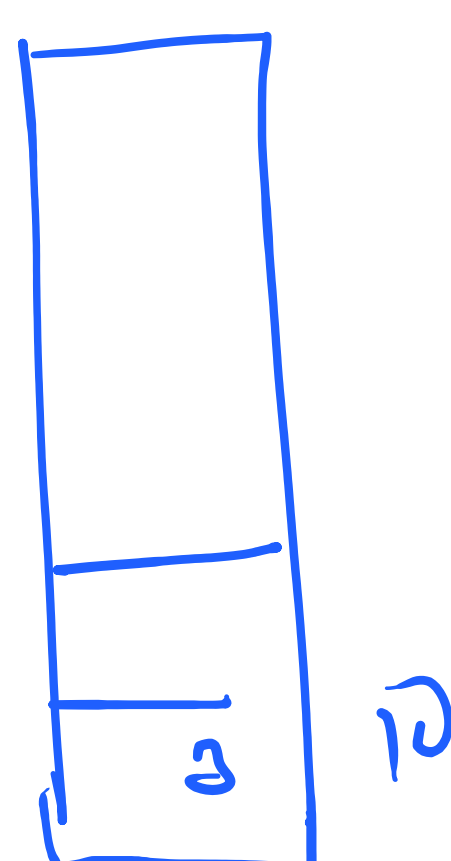


Given an array of integers, for each element in the array, find the **next greater element (NGE)** in the array. The next greater element for an element x is the first element y to the right of x in the array such that $y > x$. If no such element exists, the NGE for that element is **-1**.

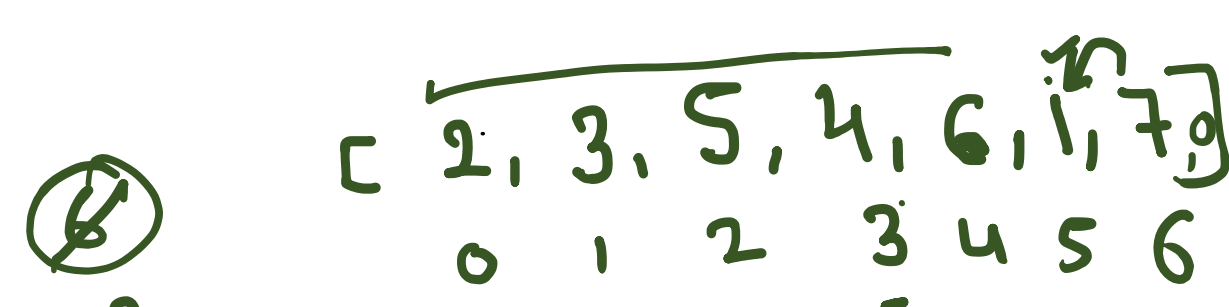


```
public static void NGE(int[] arr) {
    Stack<Integer> st = new Stack<>();
    int[] ans = new int[arr.length];
    for (int i = 0; i < arr.length; i++) {
        while (!st.isEmpty() && arr[i] > arr[st.peek()]) {
            ans[st.pop()] = arr[i];
            st.push(i);
        }
    }
    while (!st.isEmpty()) {
        ans[st.pop()] = -1;
    }
    for (int i = 0; i < ans.length; i++) {
        System.out.println(arr[i] + " " + ans[i]);
    }
}
```



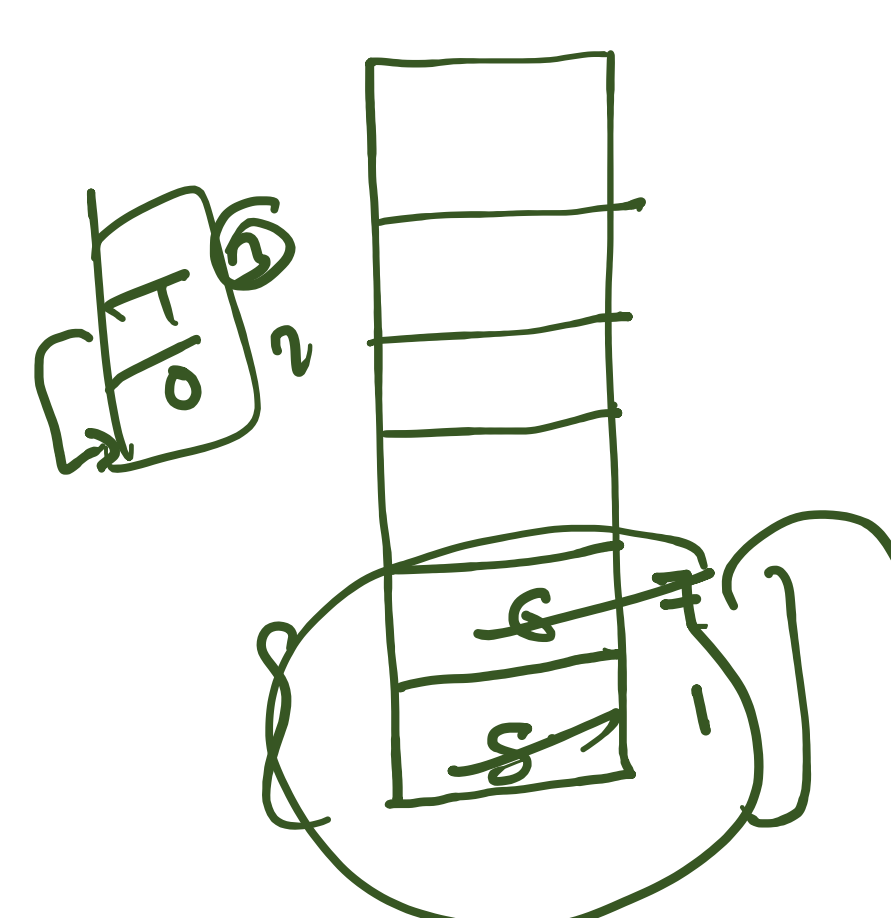
$$7 \times (7 - 5 - 1) = 7$$
$$1 \times 7 = 7$$

$$\begin{array}{r} \overline{12} \quad \overline{19} \\ \times \quad \times \\ \hline \end{array}$$



$$5 \times (3 - 1 - 1) = 5 \quad R = S \checkmark$$

```
int r=arr.length;
while(!st.isEmpty()) {
    int h=arr[st.pop()];
    if(st.isEmpty()) {
        area=Math.max(area, h*r);
    }
    else {
        int l=st.peek();
        area=Math.max(area, h*(r-l-1));
    }
}
```



$$\begin{aligned} 6 \times (5-3-1) &= 12 \\ 4 \times (5-1-1) &= 12 \\ 3 \times (5-0-1) &= 12 \\ 2 \times 5 &= 10 \end{aligned}$$

```

1  for c i = 0 i < n i++ {
2  while (!st.isEmpty() && arr[c] < arr[st.peek()])
3  {
4      R = i, h = arr[st.pop()]
5      area = max(area, h * (R - L))
6      L = st.peek()
7  }
8  st.push(c)
9  }

```

$$7 \times (7-5) = 14$$

1	0	1	0	0
1	0	1	1	1
1	1	1	1	1
1	0	0	1	0

$[1, 0, 0, 1, 0] \rightarrow 1x$
 $\downarrow$
 $[2, 1, 1, 2, 1] \rightarrow 5x$
 $[3, 0, 2, 3, 2] \rightarrow 6$
 $[4, 0, 3, 4, 0] \rightarrow 4$

